

Fernando Silva da Luz

Senior Software Engineer | Applied AI & Product Engineering

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PROFILE

Software engineer with 13+ years across medical software, computer vision and interactive products. Built production computer-vision workflows, shipped a voice-based LLM feature in a mobile AR game, and currently develop C++/Qt medical applications. Experience spans Python services, computer-vision model training, inference, client integration and product delivery.

PROFESSIONAL EXPERIENCE

Zimmer Biomet / Orthosoft

Senior Software Developer | Aug 2023 - Present

- Develop C++17/Qt medical applications for robotic-assisted surgery, contributing to medical-image processing and reconstruction workflows.
- Evaluate spatial registration methods through custom scripts and established libraries; work with VTK, ITK, Python, CMake and Conan.
- Contribute to backend integration of segmentation-model inference using models developed by other specialists; implement GitLab CI automation for inference runs and validation-data collection.
- Previously developed mixed-reality prototypes with Unity and HoloLens 2.

Prologue AI

Senior Software / Unity Developer | Jun 2022 - Jul 2023

- Took CrimeTrip, a Unity AR adventure game, from prototype through release on Android and iOS.
- Built and shipped a voice-based character interrogation feature integrating speech recognition, OpenAI API responses and ElevenLabs speech synthesis.
- Implemented pre-request rules, persona context and keyword-based output checks; validated interaction flows through manual team QA.
- Built backend services with FastAPI, Azure Functions and PlayFab, integrated with a modular, event-driven Unity client using HTTP and WebSocket communication.

BairesDev

Software Engineer, Computer Vision | Aug 2020 - Dec 2021

- Built production Python computer-vision workflows for annotation, augmentation, model training, retraining and inference, using Detectron2, OpenCV and TensorFlow.
- Combined 2D detections, 3D point-cloud data and camera geometry to estimate window dimensions and area from building-facade captures.
- Implemented capture-quality checks and coordinated image collection with the client and external contributors; built capture/detection interfaces and a Unity-React Native bridge.

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Selected work and technical profile

SELECTED PROJECTS

AI-Pulse | TypeScript / Node.js / Electron

Desktop application in development with local GGUF-model inference, a background service, REST/WebSocket APIs and SQLite.

github.com/FernandoSLuz/ai-pulse

AgentFare | Python CLI / Alpha

Alpha CLI for local analysis of coding-agent costs and rule-based model routing.

github.com/FernandoSLuz/agentfare

VEKARRA | Unity / C#

Solo tactical roguelite in development: seeded maps, active-time combat, skill drafting, saves and run-based permadeath. Upcoming on Steam.

[Steam](#)

TECHNICAL SKILLS

Languages: C++17, C#, Python, TypeScript

AI and vision: Detectron2, OpenCV, TensorFlow, OpenAI API, ElevenLabs, VTK, ITK

Services and data: FastAPI, Azure Functions, PlayFab, REST, WebSocket, SQLite

Applications: Qt 5/6, Unity, Electron; Android/iOS delivery, AR, HoloLens 2

Engineering: Git, GitLab CI, CMake, Conan; modular and event-driven software

ADDITIONAL EXPERIENCE

Founder and Tech Lead | Down Below | Sep 2021 - May 2022

Independent JRPG project: gameplay systems, production scope, art direction and marketing as a solo founder.

Earlier roles: IT Director, RSTcom; Senior Unity Developer, Expansão; Developer, TeleIDEA.

EDUCATION

Bachelor's degree in Computer Science | FMU | Completed June 2024

Game Design studies | Universidade Anhembi Morumbi | Degree not completed

LANGUAGES

Portuguese: native | English: fluent | French: intermediate

Open to relocation. US sponsorship required. Canada: employer-specific work permit.

Two weeks' notice; start date subject to work authorization.